Meng Wu
Ph.D. Candidate, Graduate Research Assistant.
Dow 825, Civil, Environmental, and Geospatial Engineering
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Objective and Intended Area of Research:

My research focuses on developing sustainable asphalt technologies that enhance pavement durability, reduce environmental impact, and promote a circular economy in transportation infrastructure, with an emphasis on high-content crumb-rubber-modified asphalt and other recycled-material-based pavement systems that enable large-scale recovery of resources from waste tires and industrial byproducts.

Education Background:

Michigan Technological University, Since August 2022
Major: Civil Engineering Supervisor: Zhanping You
GPA: 3.94/4.0
Research interests**:** Asphalt binder performance; Crumb rubber-modified asphalt; Superpave volumetric mix design and balanced mix design; Molecular simulation of asphalt material

Southeast University, September 2019 - July 2022
Major: Road and Railway Engineering Supervisor: Weiguang Zhang
GPA: 90.3/100
Master's Thesis: Multi-scale Research on Strength Formation Mechanism of Emulsified Cold Recycled Mixture based on molecular dynamics simulation

RWTH Aachen University, October 2018-April 2019
Grade: Very good
Research project: EDEM Academic- Simulating Granular Materials Behaviors

Chang'an University, August 2015- July 2019
GPA: 90/100 Rank: 2/155

Publication:

[1] **Wu, M.**, M. Li, L. Yin, and Z. You, Asphalt-rubber interaction in crumb rubber modified asphalt: a review. Journal of Cleaner Production, 2026. 546: p. 147766. DOI: 10.1016/j.jclepro.2026.147766

[2] **Wu, M.**, K. A. Boateng, L. Yin, Z. Liu, Z. You, and D. Jin (2025), High-content crumb rubber modified asphalt mixture via wet process: Laboratory evaluation and field application. Construction and Building Materials, 494: p. 143438. DOI: 10.1016/j.conbuildmat.2025.143438

[3] **Wu, M.**, L. Yin, M. Li, Z. You, D. Jin, and K. Xin (2025), A state-of-the-art review of asphalt aging behavior at macro, micro, and molecular scales. Construction and Building Materials, 2025. 460: p. 139738 DOI: 10.1016/j.conbuildmat.2024.139738

[4] **Wu, M.**, Li, M., & You, Z. (2024). Asphalt property prediction through high-throughput molecular dynamics simulation. Computer-Aided Civil and Infrastructure Engineering, 1–15. DOI: 10.1111/mice.13325

[5] **Wu, M.**, You, Z., Jin, D., Yin, L., & Xin, K. (2024). Aging effects on asphalt adhesive properties: molecular dynamics simulation of chemical composition and structural changes. Molecular Simulation, 1–19. DOI:10.1080/08927022.2024.2359568

[6] **Wu, M.** and Z. You. (2023). Molecular dynamics models to investigate the diffusion behavior of emulsified asphalt. Construction and Building Materials, 2023. 409: p. 134061. DOI:10.1016/j.conbuildmat.2023.134061

[7] **Wu, M.**, You, Z., & Jin, D. (2023). Adhesion Performance of Rubber Modified Asphalt in Chip Seal: A Molecular Dynamic Study. Materials, 16(18), 6324. DOI:10.3390/ma16186324

[8] **Wu, M.**, Xu, G., Luan, Y., Zhu, Y., Ma, T., & Zhang, W. (2022). Molecular dynamics simulation on cohesion and adhesion properties of the emulsified cold recycled mixtures. Construction and Building Materials, 333, 127403. DOI:10.1016/j.conbuildmat.2022.127403

[9] Xu, G., Yao, Y., **Wu, M.**, & Zhao, Y. (2023). Molecular simulation and experimental analysis on co-aging behaviors of SBS modifier and asphalt in SBS-modified asphalt. Molecular Simulation, 49(7), 629-642. DOI:10.1080/08927022.2023.2182134 (Corresponding author)

[10] Zhang, W., Ahmad, K. N., Tong, Z., Hu, Z., Wang, H., **Wu, M.**, ... & Mohammad, L. N. (2023). In-Time Density Monitoring of In-Place Asphalt Layer Construction via Intelligent Compaction Technology. Journal of Materials in Civil Engineering, 35(1), 04022386. DOI:10.1061/(ASCE)MT.1943-5533.0004558 (Corresponding author)

[11] Yin, L., **Wu, M.,** Liu, Z., Xin, K., Jin, D., & You, Z. (2026). Impact of extended storage and transportation on the performance and workability of high-content rubber-modified asphalt. International Journal of Pavement Engineering, 27(1), 2627429. DOI: 10.1080/10298436.2026.2627429

[12] Yin, L., Jin, D., **Wu, M.**, Liu, Z., & You, Z. (2025). Performance of high-rubber-content modified asphalt chip seal in wet-freezing environments. Journal of Cleaner Production, 519, 145993. DOI:10.1016/j.jclepro.2025.145993

[13] Xin, K.; **Wu, M.**; Jin, D.; You, Z. A Case Study of Pavement Construction Materials for Wet-Freeze Regions: The Application of Waste Glass Aggregate and High-Content Rubber Modified Asphalt. Buildings 2025, 15, 1637.DOI:10.3390/buildings15101637

[14] Yao, Y., G. Xu, M. Wu, and M. Zhao. (2023). Exploring the influence of cement and cement hydration products on strength and interfacial adhesion in emulsified cold recycled mixture: A molecular dynamics and experimental investigation. Construction and Building Materials, 409: p. 134050. DOI:10.1016/j.conbuildmat.2023.134050

[15] Zhu, Y., Ma, T., Xu, G., Fan, J., Zhang, Y., & **Wu, M.** (2023). Study of the Mixing between Asphalt and Rejuvenator in Hot In-Place Recycled Layer. Journal of Transportation Engineering, Part B: Pavements, 149(2), 04023005. DOI:10.1061/JPEODX.PVENG-1033

[16] Luan, Y., Ma, T., Wang, S., Ma, Y., Xu, G., & **Wu, M.** (2022). Investigating mechanical performance and interface characteristics of cold recycled mixture: Promoting sustainable utilization of reclaimed asphalt pavement. Journal of Cleaner Production, 369, 133366. DOI:10.1016/j.jclepro.2022.133366

[17] Fu, Y., **Wu, M.**, Hei, T., Dong, Z., Hu, J., & Zhang, W. (2022). Research on the Adhesion and Self-healing Properties of Bio-asphalt Based on Molecular Simulation. Advance Researches in Civil Engineering, 4(2), 24-43. DOI:10.30469/arce.2022.157270

[18] Zhang, W., Luan, Y., Ma, T., Wang, S., Chen, J., Li, J., & **Wu, M.** (2021). Multilevel analysis of the aging mechanisms and performance evolution of rubber-modified asphalt. Journal of Materials in Civil Engineering, 33(12), 04021365. DOI:10.1061/(ASCE)MT.1943-5533.0004000

[19] Zhang, W., Lee, J., Ahn, H. J., Le, Q., **Wu, M.**, Zhu, H., & Zhang, J. (2019). Field Investigation of Clay Balls in Full-Depth Asphalt Pavement. Materials, 12(18), 2879. DOI:10.3390/ma12182879

Academic Conference Activity:

[1] **Meng Wu**, Lei Yin, Zhongda Liu, Zhanping You. 105th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., January 2026. Poster Title: Effects of Thermal Curing on High-Content Crumb Rubber Modified Asphalt: Binder Evaluation and Field Mixture Performance.

[2] Zhongda Liu, Dongzhao Jin, Lei Yin, **Meng Wu**, Zhongqi Fan, Zhanping You. 105th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., January 2026. Poster Title: Laboratory and Field Evaluation of Soybean Oil-Based Dust Suppressants for Unpaved Roads: Performance and Durability.

[3] Dongzhao Jin, Sepehr Mohammad, Lei Yin, Zhongda Liu, **Meng Wu**, Stephen Techtmann, Zhanping You. 105th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., January 2026. Poster Title: Cold In-Place Recycling with 100% RAP Rejuvenated by Soybean Oil: Laboratory and Field Evaluation.

[4] Dongzhao Jin, Zhongda Liu, Kwadwo Ampadu Boateng, **Meng Wu**, Zhanping You. 105th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., January 2026. Poster Title: Laboratory and Field Performance of Bridge Asphalt Overlay in Wet-Freeze Climate: A Case Study of the Mackinac Bridge in Michigan.

[5] **Meng Wu**, Kwadwo Ampadu Boateng, Lei Yin, Dongzhao Jin, Kai Xin, Zhanping You. 104th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., January 2025. Poster Title: Laboratory Evaluation of Mixture Performance of High-Content Rubber Asphalt Using Wet Process.

[6] Lei Yin, Dongzhao Jin, Qi Ren, **Meng Wu**, Zhanping You. 104th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., January 2025. Poster Title: Impact of Lubricating Oil-Treated Crumb Rubber (LOCR) on Asphalt Overlay Performance.

[7] Dongzhao Jin, Sepehr Mohammadi, Kai Xin, Lei Yin, **Meng Wu**, Qi Ren, Zhanping You. 104th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., January 2025. Poster Title: Enhancing Asphalt Pavement Durability with Rubber Pellets: A Case Study in Michigan.

[8] **Meng Wu**, Miaomiao Li, Zhanping You. Association of Asphalt Paving Technologists (AAPT) Annual Meeting & Technical Sessions, Chicago, Illinois, September 2024. Poster Title: Asphalt Property Prediction Through High-Throughput Molecular Dynamics Simulation.

[9] Kai Xin, Lei Yin, **Meng Wu**, Zhanping You. 103rd Transportation Research Board (TRB) Annual Meeting, Washington, D.C., January 2024. Poster Title: Adhesion Characteristics Between Epoxy Binder and Recycled Glass.

Peer Review Service:

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